Chapter 13

Vectors I

Contents

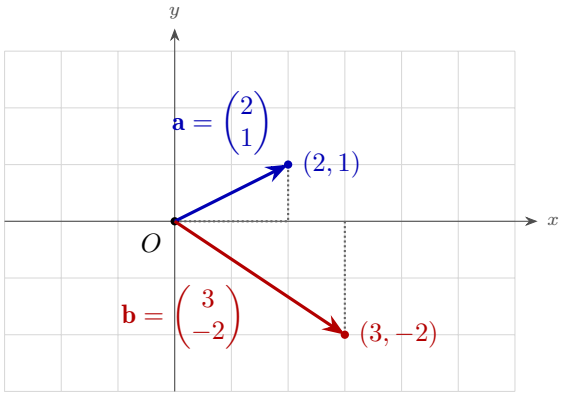
13.1 Vectors in Two Dimensions . . . 2
13.2 Vectors in Three Dimensions . . . 4
13.3 Basic Terminology and Notation . . . 6
13.3.1 Ratio Theorem . . . 8
13.4 Scalar Product . . . 9
13.4.1 Scalar (Dot) Product . . . 9
13.4.2 Basic Properties of Scalar Product . . . 9
13.4.3 Angle Between Two Vectors . . . 10
13.4.4 Perpendicular Vectors . . . 11
13.4.5 Length of Projection & Projection Vector . . . 12
13.5 Vector Product . . . 13
13.5.1 Basic Properties of Vector Product . . . 14
13.5.2 Resultant Vector of the Vector Product . . . 15
13.5.3 Parallel Vectors . . . 16
13.5.4 Vector Product Formula . . . 17
13.5.5 Application of Vector Product: Perpendicular Distance . . . 18
13.5.6 Areas of Triangles & Parallelograms . . . 19
13.5.7 Summary of Scalar & Vector Products . . . 21

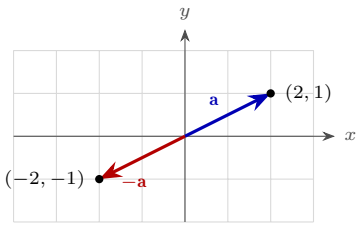
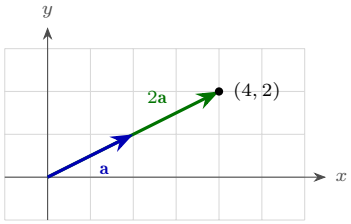
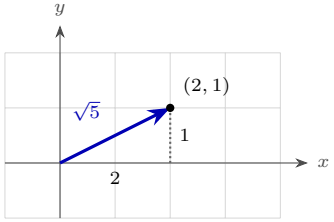
13.1 Vectors in Two Dimensions

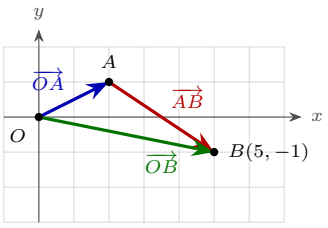
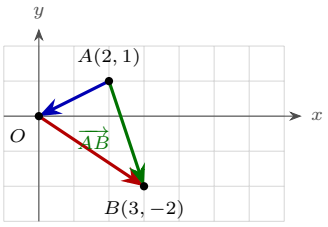
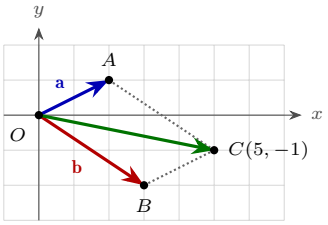
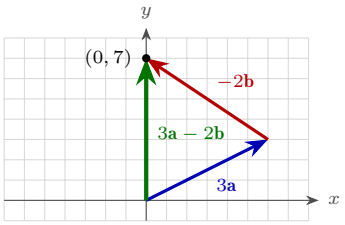
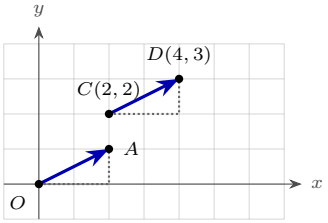
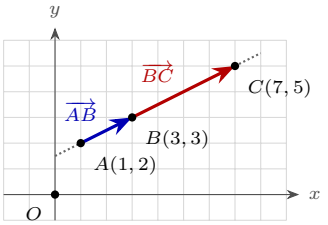
In Secondary 4, we have learned vectors in two dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$$

A vector describes a quantity that has both magnitude and direction. Vectors let us describe position, displacement and direction algebraically, so that geometric problems can be solved by calculation. In H2 Mathematics, we extend vectors from two dimensions to three dimensions, and use them to work with lines and planes, angles, distances, intersections and the foot of the perpendicular. Outside mathematics, vectors are used for forces and velocities in physics, in navigation, in computer graphics and 3D modelling, and throughout engineering. Vectors also lie at the heart of mathematics in linear algebra, which has foundational applications in fields such as physics, engineering, computer science, data science, machine learning and artificial intelligence, economics and cryptography.



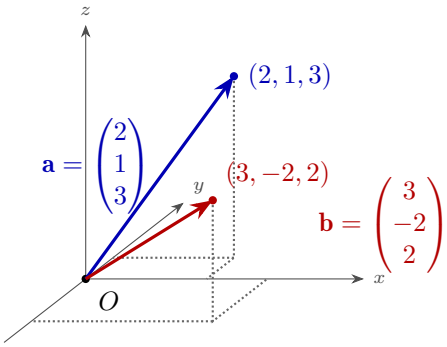
Name	Equation	Diagram
Negative of **a** $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$	
$2\mathbf{a}$	$k\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} kx \\ ky \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 4 \\ 2 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \end{pmatrix}\right\rvert = \sqrt{x^2 + y^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2 + 1^2}$ $= \sqrt{5}$	

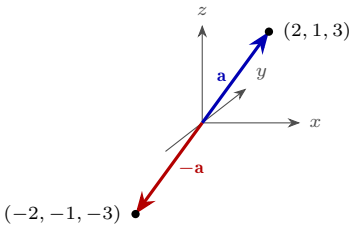
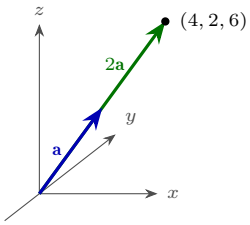
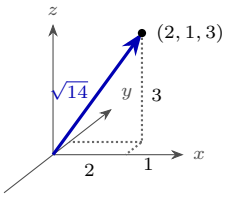
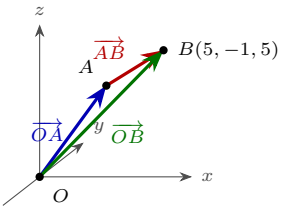
Name	Equation	Diagram
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = 3\begin{pmatrix} 2 \\ 1 \end{pmatrix} - 2\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 4-2 \\ 3-2 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\,\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 3-1 \\ 3-2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 7-3 \\ 5-3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ $= 2\,\overrightarrow{AB}$	

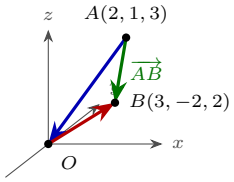
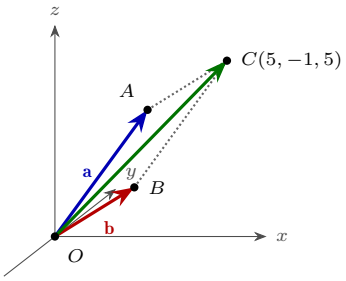
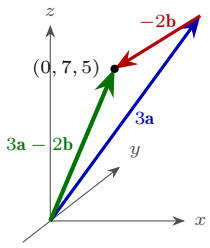
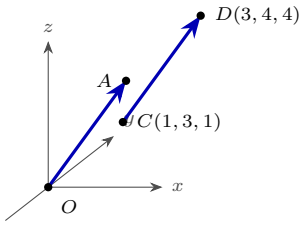
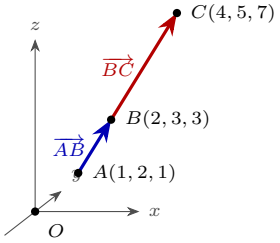
13.2 Vectors in Three Dimensions

In H2 Mathematics, we extend vectors to three dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$



Name	Equation	Diagram
Negative of **a** $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \\ -3 \end{pmatrix}$	
2a	$k\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} kx \\ ky \\ kz \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 6 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\rvert = \sqrt{x^2+y^2+z^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2+1^2+3^2}$ $= \sqrt{14}$	
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	

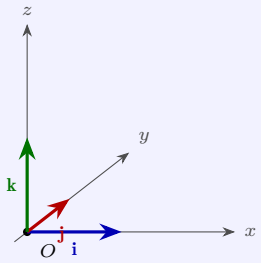
Name	Equation	Diagram
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \\ \lambda a_3 + \mu b_3 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 6 \\ 3 \\ 9 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \\ 4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 3-1 \\ 4-3 \\ 4-1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\,\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 2-1 \\ 3-2 \\ 3-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 4-2 \\ 5-3 \\ 7-3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$ $= 2\,\overrightarrow{AB}$	

13.3 Basic Terminology and Notation

Definition 13.3.1. Unit vectors i, j, k

i, **j** and **k** are the unit vectors in the positive directions of the x-, y- and z-axes:

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



Any vector can be written in terms of **i**, **j**, **k**:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Example 13.3.1

Write in column vector form: (a) $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ (b) $3\mathbf{i} + 2\mathbf{k}$. (c) Write $\begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix}$ in terms of $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Solution.

(a) $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}$ (b) $3\mathbf{i} + 2\mathbf{k} = 3\mathbf{i} + 0\mathbf{j} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$ (c) $\begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix} = 5\mathbf{i} - \mathbf{j}$

Theorem 13.3.1. Magnitude of a vector

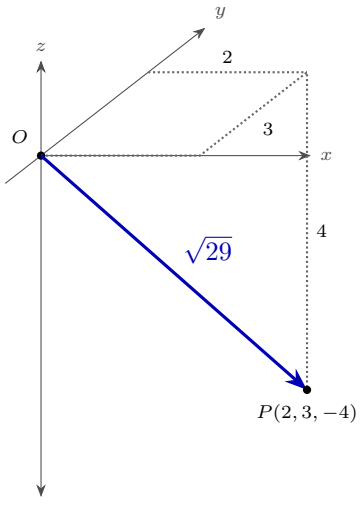
$$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad \Rightarrow \quad |\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$$

Example 13.3.2

Find $|2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}|$.

Solution.

$$\begin{aligned} |2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}| &= \sqrt{2^2 + 3^2 + (-4)^2} \\ &= \sqrt{4 + 9 + 16} \\ &= \sqrt{29} \end{aligned}$$



Theorem 13.3.2. Parallel vectors

$$\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} = \lambda\mathbf{b} \text{ for some scalar } \lambda \neq 0 \quad (\text{alternatively, } \mathbf{a} \times \mathbf{b} = \mathbf{0})$$

$\lambda > 0$: same direction; $\lambda < 0$: opposite directions.

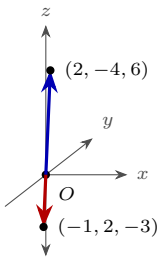
Example 13.3.3

(a) Show that $\begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$. (b) Find p and q such that $\begin{pmatrix} p \\ 6 \\ q \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$.

Solution.

(a) $$\begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} = -2 \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} \parallel \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} \text{ (shown)}$$



(b) $$\begin{pmatrix} p \\ 6 \\ q \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

$$6 = 3\lambda \Rightarrow \lambda = 2$$

$$\therefore p = 2, \quad q = -4$$

Theorem 13.3.3. Unit vectors

The unit vector in the direction of $\mathbf{a}$ is

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Example 13.3.4

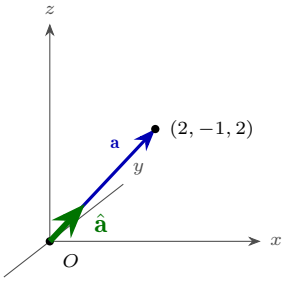
Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Find (a) the unit vector in the direction of $\mathbf{a}$, (b) the vectors of magnitude 6 parallel to $\mathbf{a}$.

Solution.

(a) $$|\mathbf{a}| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3$$

$$\hat{\mathbf{a}} = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

(b) $$\pm 6\hat{\mathbf{a}} = \pm \frac{6}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \pm \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix}$$



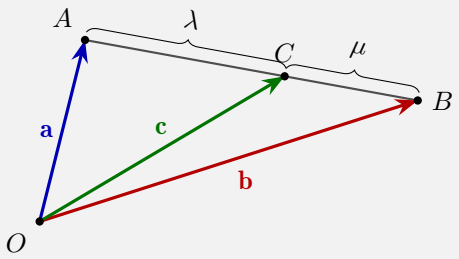
13.3.1 Ratio Theorem

Theorem 13.3.4. Ratio Theorem

If C lies on AB with $AC : CB = \lambda : \mu$, then

$$\overrightarrow{OC} = \frac{\mu\overrightarrow{OA} + \lambda\overrightarrow{OB}}{\lambda + \mu}$$

i.e. $\mathbf{c} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$.



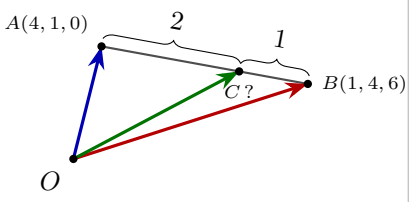
Example 13.3.5

$\overrightarrow{OA} = \begin{pmatrix}4\\1\\0\end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix}1\\4\\6\end{pmatrix}$. C lies on AB with $AC : CB = 2 : 1$. Find $\overrightarrow{OC}$.

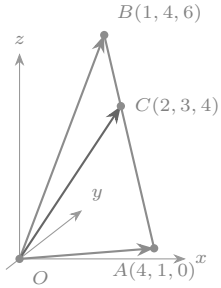
Solution.

$$\begin{aligned}\overrightarrow{OC} &= \frac{1\overrightarrow{OA} + 2\overrightarrow{OB}}{2+1}\\ &= \frac{1}{3}\left[\begin{pmatrix}4\\1\\0\end{pmatrix} + \begin{pmatrix}2\\8\\12\end{pmatrix}\right]\\ &= \frac{1}{3}\begin{pmatrix}6\\9\\12\end{pmatrix}\\ &= \begin{pmatrix}2\\3\\4\end{pmatrix}\end{aligned}$$

Diagram you can draw:



Actual diagram: (you do not need to draw this)



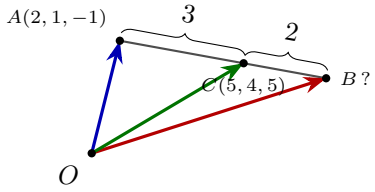
Example 13.3.6

A has position vector $\begin{pmatrix}2\\1\\-1\end{pmatrix}$ and C has position vector $\begin{pmatrix}5\\4\\5\end{pmatrix}$. C lies on AB with $AC : CB = 3 : 2$. Find $\overrightarrow{OB}$.

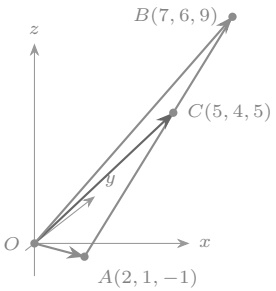
Solution.

$$\begin{aligned}\overrightarrow{OC} &= \frac{2\overrightarrow{OA} + 3\overrightarrow{OB}}{3+2}\\ \therefore\ \overrightarrow{OB} &= \frac{5\overrightarrow{OC} - 2\overrightarrow{OA}}{3}\\ &= \frac{1}{3}\left[\begin{pmatrix}25\\20\\25\end{pmatrix} - \begin{pmatrix}4\\2\\-2\end{pmatrix}\right]\\ &= \frac{1}{3}\begin{pmatrix}21\\18\\27\end{pmatrix}\\ &= \begin{pmatrix}7\\6\\9\end{pmatrix}\end{aligned}$$

Diagram you can draw:



Actual diagram: (you do not need to draw this)



13.4 Scalar Product

In this section, we learn about how we can multiply vectors together. There are two different ways we can do so, namely the Scalar Product and Vector Product. The Scalar and Vector products each have their own respective applications in the study of Vectors. We will learn about these applications more in depth later in this section.

13.4.1 Scalar (Dot) Product

Definition 13.4.1. Scalar (dot) product

The scalar or dot product of two vectors $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ is defined as

$$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_1b_1 + a_2b_2 + a_3b_3.$$

Example 13.4.1

Find $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}$.

Solution.

$$\begin{aligned} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix} &= 1(5) + 2(-1) + 3(3) \\ &= 12 \end{aligned}$$

13.4.2 Basic Properties of Scalar Product

Taking reference to the problems in the Exercise Worksheet (Section 7, Scalar Product), we can see that the Scalar Product has properties that are similar to basic algebra multiplication. In general, we have the following properties of the Scalar Product.

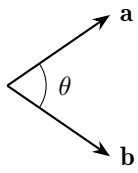
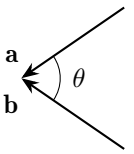
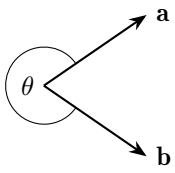
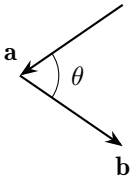
Theorem 13.4.1. Laws of scalar product

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
  $\mathbf{a} \cdot \mathbf{b} = 22 = \mathbf{b} \cdot \mathbf{a}$
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}, \quad (\mathbf{b} + \mathbf{c}) \cdot \mathbf{a} = \mathbf{b} \cdot \mathbf{a} + \mathbf{c} \cdot \mathbf{a}$
  $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = 13 = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- $\mathbf{a} \cdot (\lambda\mathbf{b}) = \lambda(\mathbf{a} \cdot \mathbf{b}) = (\lambda\mathbf{a}) \cdot \mathbf{b}, \quad \lambda \in \mathbb{R}$
  $\mathbf{a} \cdot (2\mathbf{b}) = 44 = 2(\mathbf{a} \cdot \mathbf{b}) = (2\mathbf{a}) \cdot \mathbf{b}$
- $|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a}$
  $\mathbf{a} \cdot \mathbf{a} = 14 = |\mathbf{a}|^2$

where $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -3 \\ 2 \\ 7 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 4 \\ -5 \\ -1 \end{pmatrix}$.

13.4.3 Angle Between Two Vectors

We now look at how we are able to use the Scalar Product to find the angle between two vectors. This is one of the applications of the Scalar Product. To be clear, when we say the angle θ between two vectors, we are referring to the non-reflex angle between them, where either both their start points coincide, or end points coincide.

Correct angle		Wrong angle	
			

In order to find this angle, we make use of the following formula.

Definition 13.4.2. (Definition of Scalar Product)

The scalar product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta,$$

where θ is the angle between two vectors $\mathbf{a}$ and $\mathbf{b}$.

Important Note: For nonzero $\mathbf{a}$, $\mathbf{b}$,

- $\mathbf{a} \cdot \mathbf{b} > 0 \iff \theta$ is acute;
- $\mathbf{a} \cdot \mathbf{b} < 0 \iff \theta$ is obtuse;
- $\mathbf{a} \cdot \mathbf{b} = 0 \iff \theta = 90°$.

Example 13.4.2

Find the angle between the following vectors $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$.

Solution.

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= |\mathbf{a}||\mathbf{b}| \cos\theta \\ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} &= \left| \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right| \left| \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} \right| \cos\theta \\ -2 + 4 + 3 &= \sqrt{14}\sqrt{9} \cos\theta \\ \frac{5}{3\sqrt{14}} &= \cos\theta \\ \theta &= \cos^{-1}\left(\frac{5}{3\sqrt{14}}\right) \\ \theta &= 63.5° \end{aligned}$$

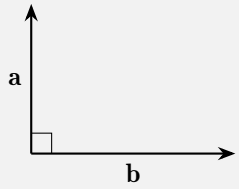
Hence we can use Definition 13.4.2 to find the non-reflex angle θ between vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}.$$

13.4.4 Perpendicular Vectors

We can see that from the previous results, that the scalar product of two non-zero vectors that are perpendicular to each other is zero.

Theorem 13.4.2. Perpendicular vectors



If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular vectors, then

$$\mathbf{a} \cdot \mathbf{b} = 0$$

Note: This result can be used to test for two perpendicular vectors, provided that the two vectors are non-zero.

Example 13.4.3

Show that the vector $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ are perpendicular to each other.

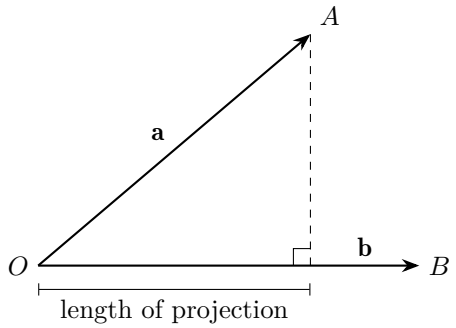
Solution.

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} \\ &= -1 + 2 - 1 \\ &= 0 \end{aligned}$$

Since $\mathbf{a} \cdot \mathbf{b} = 0$, $\mathbf{a}$ & $\mathbf{b}$ are perpendicular.

13.4.5 Length of Projection & Projection Vector

The projection vector of a vector **a** onto **b** is the vector component of **a** that is parallel to **b**. The magnitude (or length) of this vector is known as the length of projection. To help you understand this better, imagine you are shining a torchlight from a point directly above **a**. The length of projection is the length of the shadow that is cast on **b**.



In order to find the length of projection and projection vector of **a** onto **b**, we can make use of the following result.

Theorem 13.4.3. Projection of Vectors

For any two non-zero vectors **a** and **b**,

$$\text{length of projection of } \mathbf{a} \text{ onto } \mathbf{b} = |\mathbf{a} \cdot \hat{\mathbf{b}}|,$$
$$\text{projection vector of } \mathbf{a} \text{ onto } \mathbf{b} = (\mathbf{a} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}}.$$

Example 13.4.4

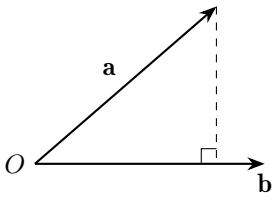
The vectors **a** and **b** are given by

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}.$$

Find the length of projection and the projection vector of the vector **a** onto **b**.

Solution.

$$\text{Length of projection} = |\mathbf{a} \cdot \hat{\mathbf{b}}|$$
$$\hat{\mathbf{b}} = \frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$$
$$\therefore \text{ length} = \left| \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \right|$$
$$= \frac{1}{3}\left| \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \right|$$
$$= \frac{1}{3} \text{ units}$$



$\therefore$ The length of projection of the vector **a** onto **b** is $\frac{1}{3}$ units. The projection vector is

$$\frac{1}{3}\left(\frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}\right) = \frac{1}{9}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$$

13.5 Vector Product

In this section, we learn how to multiply vectors together using the Vector Product. The Vector Product is slightly more difficult to apply, in comparison to the Scalar Product.

Definition 13.5.1. Vector (cross) product

The vector or cross product of two vectors $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ (denoted by $\mathbf{a} \times \mathbf{b}$) is

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$$

Example 13.5.1

Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix}$, find $\mathbf{a} \times \mathbf{b}$.

Solution.

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 2(-2) - 3(3) \\ 3(-1) - 1(-2) \\ 1(3) - 2(-1) \end{pmatrix}$$

$$\therefore \ \mathbf{a} \times \mathbf{b} = \begin{pmatrix} -13 \\ -1 \\ 5 \end{pmatrix}$$

13.5.1 Basic Properties of Vector Product

Taking reference to the problems in the Exercise Worksheet (Section 8, Vector Product), we can see that the Vector Product has properties that are similar to that of scalar product, and basic algebra. While some laws do not hold true for the vector product, others still hold true.

Theorem 13.5.1. Laws of scalar and vector product

	Scalar Product	**Vector Product**
Commutative law	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$	$\mathbf{a}\times\mathbf{b}\neq\mathbf{b}\times\mathbf{a}$
		$\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$
Distributive law	$\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$	$\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$
Associative law	$\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})$	$\lambda(\mathbf{a}\times\mathbf{b})=(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})$
	$\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$	$\mathbf{a}\times\mathbf{a}=\mathbf{0}$

In summary, the laws of the Vector Product are similar to the laws of scalar product and basic algebra. The majority of it is rather intuitive. The only peculiar results are that $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$, and that $\mathbf{a}\times\mathbf{a}=\mathbf{0}$.

Important Point: Note that applying the Vector product would yield a vector as the result, but applying the Scalar Product would yield a Scalar constant as the result.

?

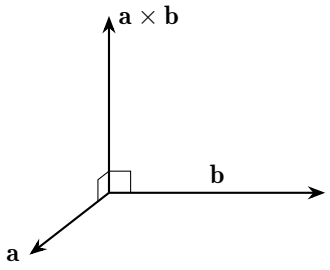
Check: Identify whether each of the following is a scalar, a vector, or is meaningless.

(1) $|\mathbf{a}|^2$,
(2) $|\mathbf{a}^2|$,
(3) $\mathbf{ab}$,
(4) $(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$,
(5) $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$,
(6) $(\mathbf{a}\cdot\mathbf{b})\times\mathbf{c}$,
(7) $(\mathbf{a}\cdot\mathbf{b})\cdot\mathbf{c}$.

Ans: (1) Scalar (2) Meaningless (3) Meaningless (4) Vector (5) Scalar (6) Meaningless (7) Meaningless.

13.5.2 Resultant Vector of the Vector Product

When we apply the Vector Product to two vectors $\mathbf{a}$ and $\mathbf{b}$, we get a resultant vector $\mathbf{a} \times \mathbf{b}$. Is there a significant relationship of the vector $\mathbf{a} \times \mathbf{b}$ to the vectors $\mathbf{a}$ and $\mathbf{b}$? There is, in fact a relationship between these three vectors.



Note: The vector $\mathbf{b} \times \mathbf{a}$ would also be perpendicular to the vectors $\mathbf{a}$ and $\mathbf{b}$, but in the opposite direction as $\mathbf{a} \times \mathbf{b}$, because $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$.

The vectors $\mathbf{a} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{a}$ are known as normal vectors to the plane in which the vectors $\mathbf{a}$ and $\mathbf{b}$ lies in.

Example 13.5.2

Given that $\mathbf{a} = 3\mathbf{i} - 5\mathbf{j} + 6\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, find a vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

Solution. A vector perpendicular to $\mathbf{a}$ and $\mathbf{b}$ is

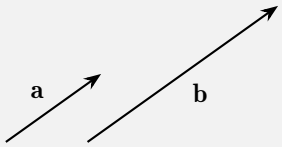
$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= \begin{pmatrix} 3 \\ -5 \\ 6 \end{pmatrix} \times \begin{pmatrix} 3 \\ 3 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} (-5)(-2) - (3)(6) \\ (6)(3) - (-2)(3) \\ (3)(3) - (3)(-5) \end{pmatrix} \\ &= \begin{pmatrix} -8 \\ 24 \\ 24 \end{pmatrix}\end{aligned}$$

? **Thinking question:** *Can you think of other vectors that are perpendicular to both* $\mathbf{a}$ *and* $\mathbf{b}$*?*

13.5.3 Parallel Vectors

We now consider the cross product of two vectors parallel to each other. Suppose **a** and **b** are two non zero vectors that are parallel to each other.

Theorem 13.5.2. Parallel vectors



If $\mathbf{a} // \mathbf{b}$,

$$\implies \mathbf{a} \times \mathbf{b} = \mathbf{0}$$

The cross product of any two non zero, parallel vectors will be equal to $\mathbf{0}$.

We consider an explicit example of the above property.

Example 13.5.3

Find $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$.

Solution.

$$\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix} = \begin{pmatrix} 6-6 \\ -3-(-3) \\ 6-6 \end{pmatrix}$$
$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Since $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, the vectors $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$ are parallel to each other.

Example 13.5.4

Show that $\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ are parallel.

Solution.

Method 1:

$$\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} = \begin{pmatrix} (-2)(2)-(1)(-4) \\ (1)(2)-(1)(2) \\ (1)(-4)-(-2)(2) \end{pmatrix}$$
$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Since **a** and **b** are non-zero vectors, **a** is parallel to **b**.

Method 2:

$$\mathbf{b} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$$
$$= 2 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$
$$= 2\mathbf{a}$$

Since **b** is a scalar multiple of **a**, **a** and **b** are parallel vectors.

Thinking question: *What do we get when we take* $\mathbf{a} \times \mathbf{a}$*? Where* $\mathbf{a}$ *is an arbitrary non zero vector.*

13.5.4 Vector Product Formula

Similar to the dot product formula $[\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta]$ the cross product has a formula as well.

Theorem 13.5.3. Vector product formula

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$$

Where θ is the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$.

Note that we have to include the modulus sign as $|\mathbf{a} \times \mathbf{b}|$ instead of $\mathbf{a} \times \mathbf{b}$, because $\mathbf{a} \times \mathbf{b}$ would return us a vector, not a scalar.

Example 13.5.5

Find the angle between the two vectors $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ by using vector product.

Solution.

$$\left| \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} \right| = \left| \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right| \left| \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} \right| \sin\theta$$

$$\left| \begin{pmatrix} 4-3 \\ -1-2 \\ 3-(-2) \end{pmatrix} \right| = \sqrt{6}\sqrt{14} \sin\theta$$

$$\left| \begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix} \right| = \sqrt{6}\sqrt{14} \sin\theta$$

$$\frac{\sqrt{35}}{\sqrt{6}\sqrt{14}} = \sin\theta$$

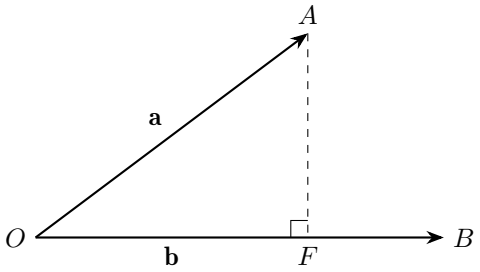
$$\theta = 40.2^\circ$$

Note: You can cross check this answer by finding the angle using the dot product.

Thinking Question: *Notice how we are able to use both dot and cross product to find the angle between any two non zero vectors. Which method is more desirable and why?*

13.5.5 Application of Vector Product: Perpendicular Distance

In this section, we deal with the perpendicular component of vectors. This concept is similar to length of projection of vectors in the previous section.



The length OF is the length of projection of $\mathbf{a}$ on $\mathbf{b}$.
The length AF is the perpendicular distance from A to the line joining O and B.

Theorem 13.5.4. Perpendicular component

For any two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$,

a) Length of vector component of $\mathbf{a}$ perpendicular to $\mathbf{b}$ (Length AF) $= |\mathbf{a} \times \hat{\mathbf{b}}|$.

b) Vector component of $\mathbf{a}$ perpendicular to $\mathbf{b} = \mathbf{a} - (\mathbf{a} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}}$.

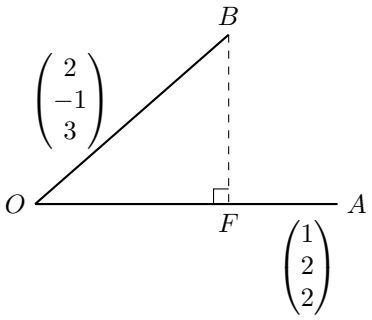
Note: You can think of this concept as the "cross product version" of the length of projection.

Example 13.5.6

Relative to an origin O, two point A and B have position vectors given by $\mathbf{a} = \mathbf{i}+2\mathbf{j}+2\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i}-\mathbf{j}+3\mathbf{k}$. Find the shortest distance from B to the line joining O and A.

Solution.

$$\begin{aligned}
\text{Required distance} &= |\mathbf{b} \times \hat{\mathbf{a}}| \\
&= \left| \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \times \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right| \\
&= \frac{1}{3} \left| \begin{pmatrix} -8 \\ -1 \\ 5 \end{pmatrix} \right| \\
&= \frac{1}{3}\sqrt{90} \\
&= \sqrt{10} \text{ units}
\end{aligned}$$

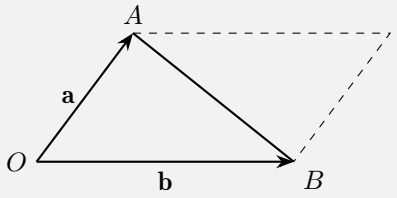


Thinking Question: *There is a difference between the shortest distance from A to OB compared to B to OA. Think about how the formula would look like for both cases. (You may draw a diagram to help illustrate this better)*

13.5.6 Areas of Triangles & Parallelograms

Another application of the Vector Product is to find areas of triangles and parallelograms.

Theorem 13.5.5. Areas



$$\text{Area of triangle } OAB = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|$$

$$\text{Area of parallelogram with sides } \mathbf{a} \text{ and } \mathbf{b} = |\mathbf{a} \times \mathbf{b}|$$

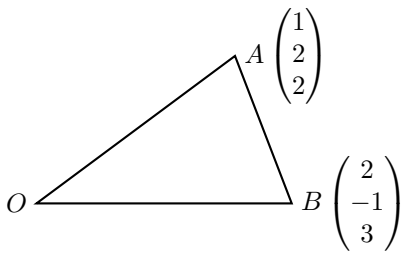
Note: We can use the area of triangle formula, $\frac{1}{2}ab\sin C$ to see why this formula holds true.

Example 13.5.7

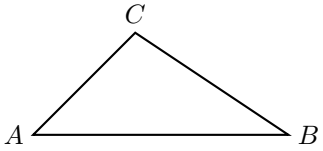
Relative to an origin O, two point A and B have position vectors given by $\mathbf{a} = \mathbf{i}+2\mathbf{j}+2\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i}-\mathbf{j}+3\mathbf{k}$. Find the area of the triangle OAB.

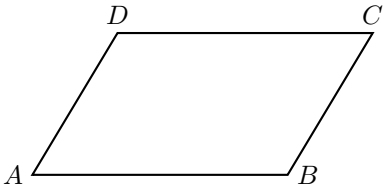
Solution.

$$\begin{aligned}
\text{Required area} &= \frac{1}{2}|\overrightarrow{OA} \times \overrightarrow{OB}| \\
&= \frac{1}{2}\left|\begin{pmatrix}1\\2\\2\end{pmatrix} \times \begin{pmatrix}2\\-1\\3\end{pmatrix}\right| \\
&= \frac{1}{2}\left|\begin{pmatrix}-8\\-1\\5\end{pmatrix}\right| \\
&= \frac{1}{2}\sqrt{90} \\
&= \frac{3}{2}\sqrt{10} \text{ units}^2
\end{aligned}$$



We can now extend this concept to find areas of triangles and parallelograms without the origin O as a vertex.





Consider the above triangle and parallelogram. The area for the respective figures can be obtained from the following result.

Theorem 13.5.6. Areas without O as a vertex

$$\text{Area of triangle } ABC = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$$

$$\text{Area of parallelogram } ABCD = |\overrightarrow{AB} \times \overrightarrow{AD}|$$

Example 13.5.8

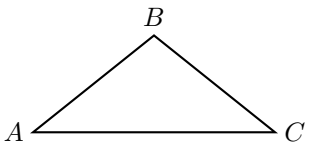
The position vectors A, B and C are $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} + 3\mathbf{k}$ and $\mathbf{c} = -\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$. Find the area of the triangle ABC.

Solution.

$$\text{Required Area} = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -1 \\ 2 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 6 \end{pmatrix}$$



$$\therefore \text{ Area} = \frac{1}{2}\left|\begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix} \times \begin{pmatrix} -2 \\ 0 \\ 6 \end{pmatrix}\right|$$

$$= \frac{1}{2}\left|\begin{pmatrix} -12 \\ -4 \\ -4 \end{pmatrix}\right|$$

$$= \frac{\sqrt{176}}{2}$$

$$= 2\sqrt{11} \text{ units}^2$$

?

Thinking Question: *If $ABCD$ is ANY quadrilateral (i.e., not necessarily a parallelogram), can you use the formula $|\overrightarrow{AB} \times \overrightarrow{AD}|$ to find the area?*
Ans: No. If the formula can be used for any quadrilateral, this would mean that any quadrilateral with adjacent sides AB and AD has the same area as the parallelogram with this pair of adjacent sides, which is not true.

13.5.7 Summary of Scalar & Vector Products

Difference between scalar and vector products

Scalar Product, $\mathbf{a} \cdot \mathbf{b}$	Vector Product, $\mathbf{a} \times \mathbf{b}$
$\mathbf{a} \cdot \mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \cos\theta$ gives a scalar	$\mathbf{a} \times \mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \sin\theta\, \hat{\mathbf{n}}$ gives a vector
Main usage: • To find angle between 2 direction vectors (also between 2 lines and between 2 planes to be discussed later) • To find length and vector of projection	Main usage: • To find a vector perpendicular to 2 given vectors • To find area of triangle and parallelogram • To find the perpendicular component of a vector with respect to another vector
For 3-D vectors, $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_1b_1 + a_2b_2 + a_3b_3$	For 3-D vector, $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$
$\mathbf{a} \cdot \mathbf{b} = 0 \implies \mathbf{a} = \mathbf{0},\ \mathbf{b} = \mathbf{0}$ or $\mathbf{a} \perp \mathbf{b}$	$\mathbf{a} \times \mathbf{b} = \mathbf{0} \implies \mathbf{a} = \mathbf{0},\ \mathbf{b} = \mathbf{0}$, or $\mathbf{a} // \mathbf{b}$
Commutative: $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$	NOT commutative: $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$ In fact, $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
Need to know: • $\mathbf{a} \cdot \mathbf{a} = \lvert\mathbf{a}\rvert^2$ • If $\mathbf{a} \perp \mathbf{b}$, then $\mathbf{a} \cdot \mathbf{b} = 0$ ($\mathbf{a}$ and $\mathbf{b}$ are non-zero vectors)	Need to know: • $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ (applies to all parallel vectors) • If $\mathbf{a}$ is parallel to $\mathbf{b}$, then $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ • The vector $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$